

IF332

FORMAL LANGUAGES

&

AUTOMATA THEORY

10
SIMPLIFICATION OF CFG

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REVIEW

Derivation Tree:

Context Free Grammar

Derivations from a Grammar

Parsing

Ambiguous Grammar

OUTLINE

Elimination of Useless Productions

Elimination of Unit Productions

Elimination of Null (ϵ) Productions

ANY PROBLEM?

- CFG_1
 - $S \rightarrow AB \mid a$
 - $A \rightarrow a$
- CFG_2
 - $S \rightarrow A$
 - $A \rightarrow B$
 - $B \rightarrow C$
 - $C \rightarrow D$
 - $D \rightarrow a \mid A$

ELIMINATION OF USELESS PRODUCTIONS

- Useless Symbols
 - Those variables or terminals that do not appear in any derivation of a terminal string from the start symbol
 - Symbols that are not generating
 - Symbols that are not reachable

ELIMINATION OF USELESS PRODUCTIONS

- CFG
 - $S \rightarrow aSa \mid Abd \mid Bde$
 - $A \rightarrow Ada$
 - $B \rightarrow BBB \mid a$
- After Elimination
 - $S \rightarrow aSa \mid Bde$
 - $B \rightarrow BBB \mid a$

ELIMINATION OF USELESS PRODUCTIONS

■ CFG

- $S \rightarrow Aa \mid B$
- $A \rightarrow ab \mid D$
- $B \rightarrow b \mid E$
- $C \rightarrow bb$
- $E \rightarrow aEa$

■ Useless Productions

- $A \rightarrow D$
- $C \rightarrow bb$
- $E \rightarrow aEa$
- $B \rightarrow E$

■ After Elimination

- $S \rightarrow Aa \mid B$
- $A \rightarrow ab$
- $B \rightarrow b$

ELIMINATION OF USELESS PRODUCTIONS

- CFG
 - $S \rightarrow aAb \mid cEB$
 - $A \rightarrow dBE \mid eeC$
 - $B \rightarrow ff$
 - $C \rightarrow ae$
 - $D \rightarrow h$
- After Elimination
 - $S \rightarrow aAb$
 - $A \rightarrow eeC$
 - $C \rightarrow ae$

ELIMINATION OF USELESS PRODUCTIONS

- CFG
 - $S \rightarrow aB$
 - $A \rightarrow bcD \mid dAC$
 - $B \rightarrow e \mid Ab$
 - $C \rightarrow bCb \mid adF \mid ab$
 - $F \rightarrow cFB$
- After Elimination
 - $S \rightarrow aB$
 - $B \rightarrow e$

ELIMINATION OF USELESS PRODUCTIONS

- CFG
 - $S \rightarrow aBD$
 - $B \rightarrow cD \mid Ab$
 - $D \rightarrow ef$
 - $A \rightarrow Ed$
 - $F \rightarrow dc$
- After Elimination
 - $S \rightarrow aBD$
 - $B \rightarrow cD$
 - $D \rightarrow ef$

ELIMINATION OF UNIT PRODUCTIONS

- Unit Productions
 - Those of the form $A \rightarrow B$ for variables A and B

ELIMINATION OF UNIT PRODUCTIONS

- CFG
 - $S \rightarrow Sb \mid C$
 - $C \rightarrow D \mid ef$
 - $D \rightarrow dd$
- After Elimination
 - $S \rightarrow Sb \mid dd \mid ef$
 - $C \rightarrow dd \mid ef$
 - $D \rightarrow dd$

ELIMINATION OF UNIT PRODUCTIONS

- CFG

- $S \rightarrow A \mid Aa$
- $A \rightarrow B$
- $B \rightarrow C \mid b$
- $C \rightarrow D \mid ab$
- $D \rightarrow b$

- After Elimination

- $S \rightarrow ab \mid b \mid Aa$
- $A \rightarrow ab \mid b$
- $B \rightarrow ab \mid b$
- $C \rightarrow b \mid ab$
- $D \rightarrow b$

ELIMINATION OF UNIT PRODUCTIONS

- CFG

- $S \rightarrow Cba \mid D$
- $A \rightarrow bbC$
- $B \rightarrow Sc \mid ddd$
- $C \rightarrow eA \mid f \mid C$
- $D \rightarrow E \mid SABC$
- $E \rightarrow gh$

- After Elimination

- $S \rightarrow Cba \mid gh \mid SABC$
- $A \rightarrow bbC$
- $B \rightarrow Sc \mid ddd$
- $C \rightarrow eA \mid f$
- $D \rightarrow gh \mid SABC$
- $E \rightarrow gh$

ELIMINATION OF NULL (ε) PRODUCTIONS

- Null (ε) Productions
 - Those of the form $A \rightarrow \varepsilon$ for some variable A
 - A variable A is nullable if $A \xRightarrow{*} \varepsilon$
 - If A is nullable, then whenever A appears in a production body, A might (or might not) derive ε

ELIMINATION OF NULL (ε) PRODUCTIONS

- CFG

- $S \rightarrow bcAd$
- $A \rightarrow \varepsilon$

- After Elimination

- $S \rightarrow bcd$

- CFG

- $S \rightarrow bcAd$
- $A \rightarrow bd \mid \varepsilon$

- After Elimination

- $S \rightarrow bcAd \mid bcd$
- $A \rightarrow bd$

ELIMINATION OF NULL (ε) PRODUCTIONS

- CFG
 - $S \rightarrow Ab \mid Cd$
 - $A \rightarrow d$
 - $C \rightarrow \varepsilon$

- After Elimination
 - $S \rightarrow Ab \mid d$
 - $A \rightarrow d$

ELIMINATION OF NULL (ϵ) PRODUCTIONS

- CFG
 - $S \rightarrow dA \mid Bd$
 - $A \rightarrow bc \mid \epsilon$
 - $B \rightarrow c$
- After Elimination
 - $S \rightarrow dA \mid d \mid Bd$
 - $A \rightarrow bc$
 - $B \rightarrow c$

ELIMINATION OF NULL (ε) PRODUCTIONS

- CFG

- $S \rightarrow AaCD$
- $A \rightarrow CD \mid AB$
- $B \rightarrow b \mid \varepsilon$
- $C \rightarrow d \mid \varepsilon$
- $D \rightarrow \varepsilon$

- After Elimination

- $S \rightarrow AaC \mid aC \mid Aa \mid a$
- $A \rightarrow C \mid AB \mid A \mid B$
- $B \rightarrow b$
- $C \rightarrow d$

ELIMINATION OF NULL (ϵ) PRODUCTIONS

- CFG
 - $S \rightarrow AB$
 - $A \rightarrow aAA \mid \epsilon$
 - $B \rightarrow bBB \mid \epsilon$

- After Elimination
 - $S \rightarrow AB \mid A \mid B$
 - $A \rightarrow aAA \mid aA \mid a$
 - $B \rightarrow bBB \mid bB \mid b$

SIMPLIFICATION OF CFG

- Convert any CFG G into an equivalent CFG that has no useless symbols, unit productions, and ε -productions
- Some care must be taken in the order of application of the constructions
- A safe order is:
 1. Eliminate ε -productions
 2. Eliminate unit productions
 3. Eliminate useless productions / symbols
- We must order the three steps above as shown, or the result might still have some of the features we thought we were eliminating

SIMPLIFICATION OF CFG

Eliminate useless productions, unit productions, and ε -productions from the CFG below.

$$S \rightarrow AA \mid C \mid bd$$

$$A \rightarrow Bb \mid \varepsilon$$

$$B \rightarrow AB \mid d$$

$$C \rightarrow de$$

1. Elimination of ε -productions

- $S \rightarrow A \mid AA \mid C \mid bd$
- $A \rightarrow Bb$
- $B \rightarrow B \mid AB \mid d$
- $C \rightarrow de$

2. Elimination of unit productions

- $S \rightarrow Bb \mid AA \mid de \mid bd$
- $A \rightarrow Bb$
- $B \rightarrow AB \mid d$
- $C \rightarrow de$

3. Elimination of useless productions

- $S \rightarrow Bb \mid AA \mid de \mid bd$
- $A \rightarrow Bb$
- $B \rightarrow AB \mid d$



PRACTICE

EXERCISES

1. Eliminate useless productions from the CFG below.

$$S \rightarrow AB \mid CA$$

$$A \rightarrow a$$

$$B \rightarrow BC \mid AB$$

$$C \rightarrow aB \mid b$$

2. Eliminate useless productions from the CFG below.

$$S \rightarrow aS \mid A \mid C$$

$$A \rightarrow a$$

$$B \rightarrow aa$$

$$C \rightarrow aCb$$

EXERCISES

3. Eliminate unit productions from the CFG below.

$$S \rightarrow Aa \mid B$$

$$A \rightarrow a \mid bc \mid B$$

$$B \rightarrow A \mid bb$$

4. Eliminate unit productions from the CFG below.

$$S \rightarrow AaC \mid aC \mid Aa \mid a$$

$$A \rightarrow C \mid AB \mid A \mid B$$

$$B \rightarrow b$$

$$C \rightarrow d$$

EXERCISES

5. Eliminate ε -productions from the CFG below.

$$S \rightarrow aAb$$

$$A \rightarrow aAb \mid \varepsilon$$

6. Eliminate ε -productions from the CFG below.

$$S \rightarrow ABaC$$

$$A \rightarrow BC$$

$$B \rightarrow b \mid \varepsilon$$

$$C \rightarrow D \mid \varepsilon$$

$$D \rightarrow d$$

EXERCISES

7. Eliminate ε -productions from the CFG below.

$$S \rightarrow AaB \mid aaB$$

$$A \rightarrow \varepsilon$$

$$B \rightarrow bbA \mid \varepsilon$$

8. Eliminate ε -productions from the CFG below.

$$S \rightarrow AB$$

$$A \rightarrow aA \mid abB \mid aCa$$

$$B \rightarrow bA \mid BB \mid \varepsilon$$

$$C \rightarrow \varepsilon$$

$$D \rightarrow dB \mid BCB$$

EXERCISES

9. Eliminate useless productions, unit productions, and ε -productions from the CFG below.

$$S \rightarrow a \mid aA \mid B \mid C$$

$$A \rightarrow aB \mid \varepsilon$$

$$B \rightarrow Aa$$

$$C \rightarrow cCD$$

$$D \rightarrow ddd$$

EXERCISES

10. Eliminate useless productions, unit productions, and ε -productions from the CFG below.

$$S \rightarrow ASB \mid \varepsilon$$

$$A \rightarrow aAS \mid a$$

$$B \rightarrow SbS \mid A \mid bb$$

REFERENCES

- Utdirartatmo, F. (2005), Teori Bahasa dan Otomata (edisi ke-2), Graha Ilmu.
- Hopcroft, John E., Rajeev Motwani, and Jeffrey D. Ullman (2006), Introduction to Automata Theory, Languages, and Computation (3rd edition), Addison-Wesley.

NEXT

Chomsky Normal Form (CNF):

Chomsky Normal Form (CNF)

Conversion of CFG to CNF

VISION

To become an **outstanding** undergraduate Computer Science program that produces **international-minded** graduates who are **competent** in software engineering and have **entrepreneurial spirit** and **noble character**.

MISSION

1. To conduct studies with the best technology and curriculum, supported by professional lecturer
2. To conduct research in Informatics to promote science and technology
3. To deliver science-and-technology-based society services to implement science and technology



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